
Assessing the Validity of Objective Evaluation Metrics for Symbolic Music Generation

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Abstract

In symbolic music generation literature, several studies have documented discrepancies between objective metrics and human evaluation [Copet et al., 2023, Wang et al., 2025]. To investigate why these occur, three test sets based on Johann Sebastian Bach chorales are evaluated with three metric families: commonly used MusPy-based distributional metrics, rule-based metrics predicated on conventions in historical counterpoint, and model likelihood. The metrics are tested on real Bach chorales, a set of shuffled Bach chorales that is created to test the metrics on their ability to distinguish sequential continuity, and a set of chorales generated with a small transformer trained on real Bach. Distributional metrics cannot distinguish real Bach from the shuffled set, and have a relatively small effect size when comparing real Bach to the generated set. Rule-based metrics, by contrast, show strong discriminative power. As each metric is blind to failures outside its own domain, single-metric evaluation is insufficient for assessing musical quality.

1 Introduction

This assignment for the course *Generative Artificial Intelligence* was written by Fabian Lohmann (student ID: 852760749). The topic is the evaluation of objective metrics used in symbolic music generation. This relates to cluster 3 (music) of the course material. The learning goals in this cluster include identifying appropriate representations of audio data, describing training strategies, and explaining the loss functions used in generative audio models, all of which are addressed in this assignment.

In recent work, Copet et al. [2023] note that in their MusicGen model a discrepancy exists between the objective Fréchet Audio Distance (FAD) score, and the human evaluation. The smallest model achieved the best FAD while receiving the lowest human ratings, suggesting that once a certain quality threshold is achieved, improvements in generated audio may deteriorate the FAD score, which the authors attribute to noisy samples in the MusicCaps reference set. Moreover, when melody conditioning is added, this degrades all the objective metrics, while keeping human benchmarks more or less the same.

While MusicGen is based on audio waveforms, the same discrepancy between objective and human evaluation can be found in symbolic music generation. In the NotaGen paper, Wang et al. [2025] note that the perplexity (PPL) score worsened after optimizing their model, whereas human evaluation showed a higher preference for the optimized version. The authors suggest that this is due to limitations of traditional language model metrics in assessing musical quality.

Furthermore, in the ProGress paper, Ni-Hahn et al. [2025] claim that in human evaluation, ProGress was preferred over Bach himself, with Bach having a substantially higher proportion of perceived “weirdness” in his music, even when evaluated by experts. Considering that Bach is one of the most

highly regarded composers in history, this further underlines the complexity of evaluating the quality of generated music, and calls into question whether even human evaluation provides a reliable gold standard.

This study investigates whether standard objective metrics for symbolic music generation actually measure musical quality, and what aspects of music they may be blind to. To do so, a small model is trained on Bach chorales using the JSB Chorales dataset [Allan and Williams, 2005], with the standard split introduced by Boulanger-Lewandowski et al. [2012]. This is then evaluated using commonly used metrics in symbolic music generation, as well as with rule-based metrics based on conventions in historical counterpoint [Huron, 2001], introduced in this study.

2 Goal

The main goal of this study is to assess whether commonly used objective evaluation metrics used in symbolic music generation literature actually measure musical quality, and what they may be blind to.

To achieve this, several sub-objectives are adopted:

1. **Training a model:** A small model is trained on Bach chorales which produces a set of generated chorales for evaluation.
2. **Constructing a controlled baseline:** By randomly reordering segments of Bach chorales, a shuffled test set is created that is locally identical to Bach, but structurally incoherent. This can be used as a negative control to test the evaluation of structural quality.
3. **Evaluating with three metric families:** Distributional metrics (based on MusPy), rule-based metrics based on counterpoint convention, and model likelihood (PPL) are applied on all three test sets (real Bach, shuffled Bach, generated).
4. **Interpreting the results:** The outcomes of the evaluation are analyzed for how well each metric assesses musical quality.

The remainder of the report is structured as follows: Section 3 describes the dataset in more depth, Section 4 covers the design methodology and implementation of this study, Section 5 presents the results of the evaluation and Section 6 discusses conclusions and limitations.

3 Data analysis

3.1 Dataset description

Chorales are melodies from the Lutheran church tradition. Johann Sebastian Bach composed four-part harmonizations around these hymns. The original melody was sung by the soprano voice, with three accompanying parts (the alto, the tenor and the bass) composed around it, following the conventions of tonal counterpoint (the practice of writing multiple simultaneous melodies according to established harmonic rules). Due to the relatively large number of harmonizations Bach composed, his chorales have long been established as a standard reference for music theorists [Allan and Williams, 2005]. Because Bach chorales have well-documented rules that govern their compositional style, they are uniquely suited to being assessed on such conventions.

The model used in this study is trained on the JSB Chorales dataset¹, which contains 382 four-part chorales composed by Johann Sebastian Bach. The original train/test split was introduced by Allan and Williams [2005], and Boulanger-Lewandowski et al. [2012] further split the test set into validation and test subsets. This resulted in a distribution of 229 training chorales, 76 validation chorales and 77 test chorales.

The JSB Chorales dataset is available at multiple temporal resolutions (quarter-note, eighth-note or sixteenth-note). A sixteenth-note quantization was chosen to preserve full rhythmic detail of the original chorales. The dataset contains a dictionary already split into ‘train’, ‘valid’ and ‘test’ keys, with each containing a list of sequences. Each sequence represents a chorale as a list of timesteps,

¹<https://github.com/czhuang/JSB-Chorales-dataset>

with every timestep containing four numbers, each representing one of the SATB (soprano, alto, tenor, bass) voices.

3.2 Data processing

In order to train the model, the data is first tokenized. Every chorale starts with a ‘beginning of sequence’ (BOS) token. Subsequently, every timestep is placed sequentially into the token list. Because every chorale is subdivided into four vocal parts, each timestep takes up four positions. Every chorale ends with an ‘end of sequence’ (EOS) token.

Across all chorales, a tonal range of 46 distinct pitches spanning 45 semitones (MIDI 36-81, or C2-A5 in musical notation) is observed. These are assigned to tokens 1 to 46. Token 0 is assigned to rests (timesteps where a voice is silent), and tokens 47 and 48 to sequence boundaries BOS and EOS. This results in a vocabulary of 49 tokens. An advantage of this method is that the token list is easily decoded for analysis with rule-based metrics.

3.3 Data cleaning

The dataset is already cleaned as-is. There are no missing or corrupted chorales in the dataset. A minority of timesteps contain fewer than four voices. These are assigned REST tokens during tokenization. No further data augmentation was applied.

3.4 Data visualization and interpretation

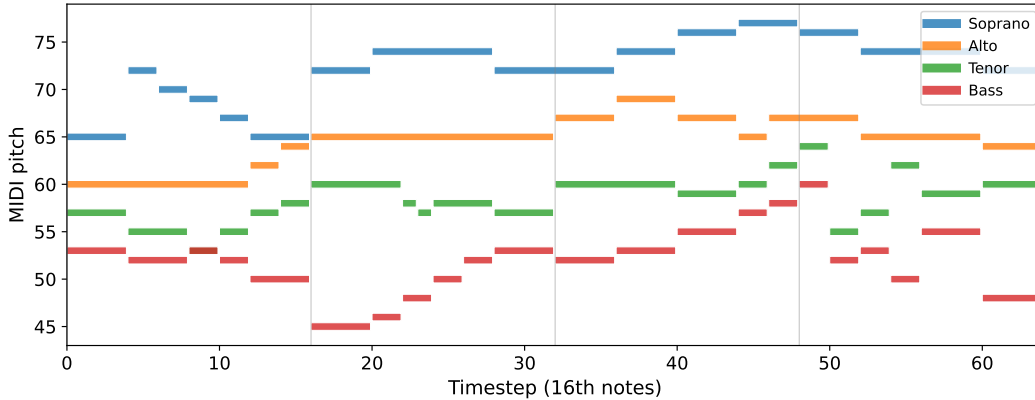


Figure 1: Piano roll of four bars from a Bach chorale in the test set. Each color represents one of the four SATB voices. A hierarchical separation between voice registers is visible.

Figure 1 shows what an excerpt of a Bach chorale looks like when visualized as a piano roll. An important characteristic of chorales is that every voice type is separated by pitch. Figure 2 illustrates the presence of this separation in Bach’s compositions. It is convention that voices maintain their registral hierarchy in accordance with SATB (soprano > alto > tenor > bass). Because Bach rarely crossed voices, this provides a ground truth for evaluating whether a model maintains chorale conventions, and is measured with the voice-crossing metric introduced in Section 4.1.

4 Methodology and Implementation

4.1 Research methodology

Three sets of data are used for testing the selection of evaluation metrics:

1. **Real Bach:** The 77 test chorales withheld from training are used as the reference standard.
2. **Shuffled Bach:** The 77 Bach chorales from the test set with randomly reordered segments. These are used as a negative control to test whether the metrics can evaluate the structural quality of music.

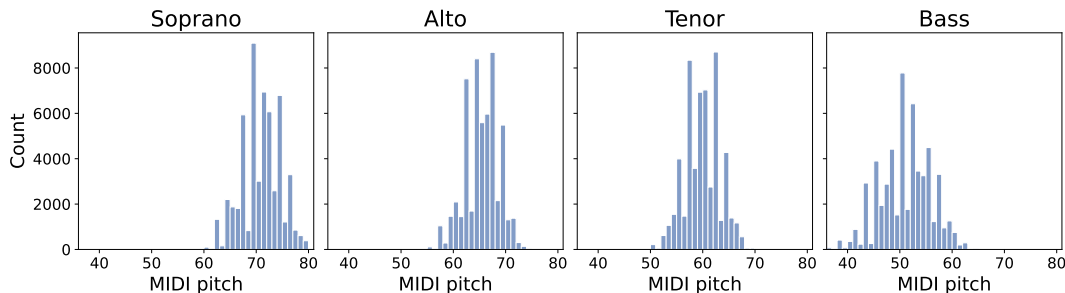


Figure 2: Pitch distribution per voice in the JSB Chorales training set. Each voice occupies a distinct range of semitones: soprano (MIDI 60–81, median 71), alto (52–74, median 65), tenor (43–69, median 59), and bass (36–66, median 50).

The chorales are reordered in 16-step segments, which would coincide with a single bar in 4/4 time signature pieces. Although not all chorales adhere to the 4/4 time signature, this does not affect the validity of the comparison, as the shuffling of fixed-length segments would affect sequential continuity regardless.

Any remainder final segments smaller than 16 steps are kept at the end of the shuffled chorales.

3. **Generated:** To match the amount of real Bach chorales that are being evaluated, 77 chorales are generated with the trained model. These are used to test whether the metrics can distinguish the qualitative differences between real Bach chorales and generated chorales.

Three families of metrics are being compared in this study:

1. **Distributional metrics** (MusPy based): These are standard metrics widely adopted in symbolic music generation literature, computed with the MusPy toolkit [Dong et al., 2020], which implements distributional metrics that measure the statistical properties of music pieces, originally proposed in earlier work [Mogren, 2016, Dong et al., 2018]. Dong et al. [2020] explicitly frame the metrics as a way to evaluate a music generation system by comparing statistical differences between training data and generated samples.

The following four metrics were selected as they are among the most frequently reported in symbolic music generation literature, for instance by Zhu et al. [2024] and Su et al. [2025].

- *Pitch class entropy*: Measures how evenly all twelve pitch classes in an octave (C, C#, D, ..., B) are distributed. A higher pitch class entropy indicates more variety in pitch.
- *Scale consistency*: The fraction of notes in a chorale that belong to a single scale. Values lower than 1.0 indicate that every note does not fit into a single key.
- *Polyphony*: The average number of notes that are simultaneously present per timestep. Since there are four voices, the maximum value is 4.0.
- *Pitch range*: The distance in semitones between the lowest and highest note in the piece among all voices.

2. **Rule-based metrics**: Whereas distributional metrics are based on surface statistics, these metrics are based on structural counterpoint principles inherent to Bach chorales. In his study on perceptual principles, Huron [2001] argues that conventions such as avoiding parallel fifths and voice crossing exist to preserve the perceptual independence of simultaneous melodic lines. These metrics were chosen to directly test whether distributional metrics miss quality differences that strict compositional conventions of a specific style, such as Bach chorales, can detect.

- *Voice crossing*: In chorales, the pitch of each voice should ideally be hierarchical at each timestep. This metric measures how often the voice registers are crossed.
- *Parallel fifths*: When two voices move in parallel at a perfect fifth (7 semitones) or an octave/unison during consecutive notes, they lose their perceptual independence as separate melodies [Huron, 2001]. Because Bach avoided this meticulously, this

metric measures how often two voices maintain a perfect fifth or an octave/unison interval when they move to consecutive timesteps.

- *Proper cadence*: In Bach chorales, the final chord is often a root-position major or minor triad, with the bass on the chord root, which gives a sense of resolution to the piece. This metric counts in how many chorales of a set this is the case.
- *Segment-boundary interval analysis*: Because in the set of shuffled chorales the connections between segments are broken, the structural cohesiveness should no longer be intact. To test for musical continuity, an analysis is performed that quantifies the jumps in pitch at segment boundaries for the soprano voice.

3. **Model likelihood** (Negative Log-Likelihood / perplexity): Negative Log-Likelihood (NLL) measures how poorly a probabilistic model fits the true data distribution. This is calculated by feeding a sequence of tokens into the model, and comparing the model’s predictions to the actual next token. Higher NLL indicates the model finds the sequence less predictable. Because shuffling the chorales preserves local structure while disrupting sequential structure, higher NLL for the shuffled pieces should indicate that the model learned structural cohesiveness. Perplexity (PPL) is the exponential of NLL, and is often used as a more intuitive representation of the model likelihood, as it directly represents how many options the model is choosing between at each step.

Model likelihood has been the primary benchmark for sequence models on the JSB Chorales dataset since Boulanger-Lewandowski et al. [2012] introduced the current split. It was also used by Huang et al. [2018] when training Music Transformer on the same corpus, and remains in use in recent work such as NotaGen [Wang et al., 2025], where its divergence from human evaluation motivates this study.

Model likelihood is not calculated for the generated chorales, as these have been generated by the model itself. Scoring generated chorales would not reveal anything because the model would always score its own samples as likely.

To validate the statistical significance of the results, Mann–Whitney U tests are applied on all continuous metrics. Mann–Whitney U is used because it does not assume the data is normally distributed [Mann and Whitney, 1947], which is appropriate here because the metric values are heavily skewed. Real Bach is compared to the generated chorales, and to the shuffled chorales. Fisher’s exact test [Fisher, 1935] is used for the cadence quality metric, because it is a binary metric.

Because 18 tests are performed in total at a threshold of $\alpha = 0.05$, the probability of at least one false positive can rise to $1 - 0.95^{18} \approx 0.60$. For this reason, the Holm–Bonferroni procedure [Holm, 1979] is applied, which corrects all p -values so that the combined risk of a false positive across all results remains at 5%.

Additionally, to measure effect size, the rank-biserial correlation [Wendt, 1972] is calculated from all Mann–Whitney U tests. The rank-biserial correlation measures how completely two groups are separated, and is calculated from the U statistic as $r_{rb} = 1 - 2U/(n_1 n_2)$, with $|r_{rb}| = 0$ indicating full overlap and $|r_{rb}| = 1$ full separation. This distinguishes metrics that detect a statistically reliable but small difference between sets from metrics that separate the sets almost entirely, and can serve as an indicator of how reliably a metric separates the two sets in pairwise comparison. For instance, at a rank-biserial correlation of $r = 0.50$, a randomly drawn chorale from one set will score higher than a randomly drawn chorale from the other set 75% of the time, as $(1 + r)/2 = (1 + 0.50)/2 = 0.75$.

4.2 Design elaboration

To generate the chorales, a small decoder-only transformer is trained using autoregressive next-token prediction. The hyperparameters are chosen using common conventions for small transformers, enabling the model to be trained locally on 4GB of VRAM: 4 layers, 4 heads, 192 dimensions, resulting in a model with ~1.85M parameters. An advantage of this model size is that it prevents the model from simply memorizing the relatively small amount of chorales, rather than generalizing.

A context window of 256 tokens is used. This corresponds to ~4 bars of music. A limitation of this is that the model cannot learn structure beyond this window. The model is trained with AdamW optimizer, which adds weight decay to prevent overfitting, with a learning rate of $3e-4$.

No further model optimization is performed, as the focus of this study is metric evaluation. A flawed model is required to accentuate the differences between the metrics when comparing gen-

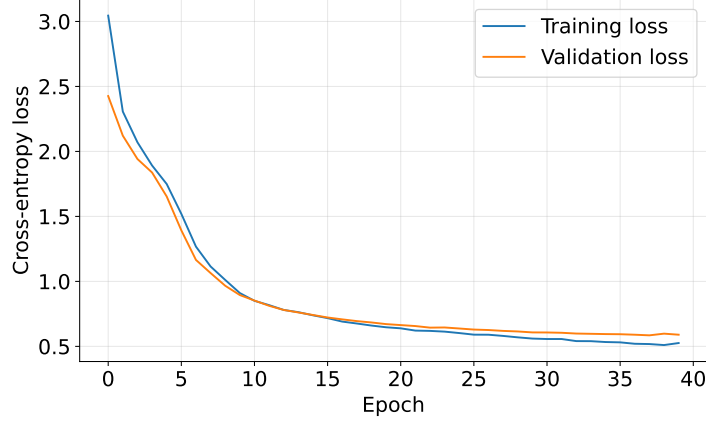


Figure 3: Training and validation loss over 40 epochs. After epoch 15 the training loss keeps going down while the validation loss flattens, suggesting some mild overfitting. The checkpoint with the lowest validation loss is used for all generation.

erated chorales to real Bach. A flawless model would leave the metrics nothing to detect, and the comparison would be uninformative. Figure 3 shows that the model converges within 40 epochs, with mild overfitting consistent with the small dataset size and the choice not to further optimize.

4.3 Implementation choices

The decoder-only architecture chosen for the model is consistent with other music generation models such as Music Transformer [Huang et al., 2018], MusicGen [Copet et al., 2023], and NotaGen [Wang et al., 2025], but applied at a smaller scale here. As with large language models, sequential prediction requires that the model can never see the future during training. This is achieved by applying a causal mask over PyTorch’s `TransformerEncoder` function, restricting each token to only attend to preceding positions.

Because the context window is limited to 256 tokens, a sliding window with stride 128 is used to create overlapping chunks, maximizing the amount of training examples from the limited amount of data.

During initial training rounds, the model often failed to include the EOS token in generated chorales, to signify the end of the piece. To resolve this, the final chunk of each Bach chorale is included at a random position to make sure EOS as well as cadences appear in the training data without positional bias. Secondly, the maximum generation length of generated chorales is set to 1600 tokens, or ~25 bars of generated music to give the model sufficient time to finish the piece.

Chorales are generated using multinomial sampling at temperature 1.0, without top-k or nucleus truncation. Temperature is deliberately not tuned so that the evaluated samples reflect the model’s learned distribution rather than a sharpened version that may suppress the errors that rule-based metrics measure.

5 Evaluation and Results

5.1 Evaluation mechanism and criteria

To assess how well each metric can measure musical quality, they are evaluated on their discriminative power between the datasets. Discriminative power is judged on both significance (Mann–Whitney U , Fisher’s exact test for cadence; both at $\alpha = 0.05$) and effect size (rank-biserial correlation). A metric that measures musical quality should (a) distinguish real Bach from shuffled chorales, which are structurally broken, and (b) distinguish real Bach from generated chorales, which are locally flawed.

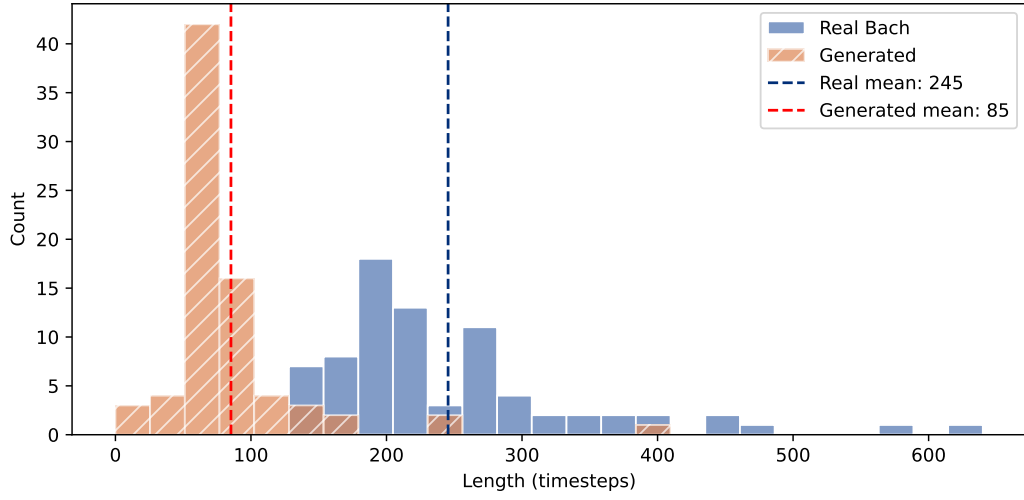


Figure 4: Histogram of chorale lengths for real Bach and generated sets. Real Bach chorales are overall longer than the generated chorales, with a mean of 245 vs. 85 timesteps, respectively.

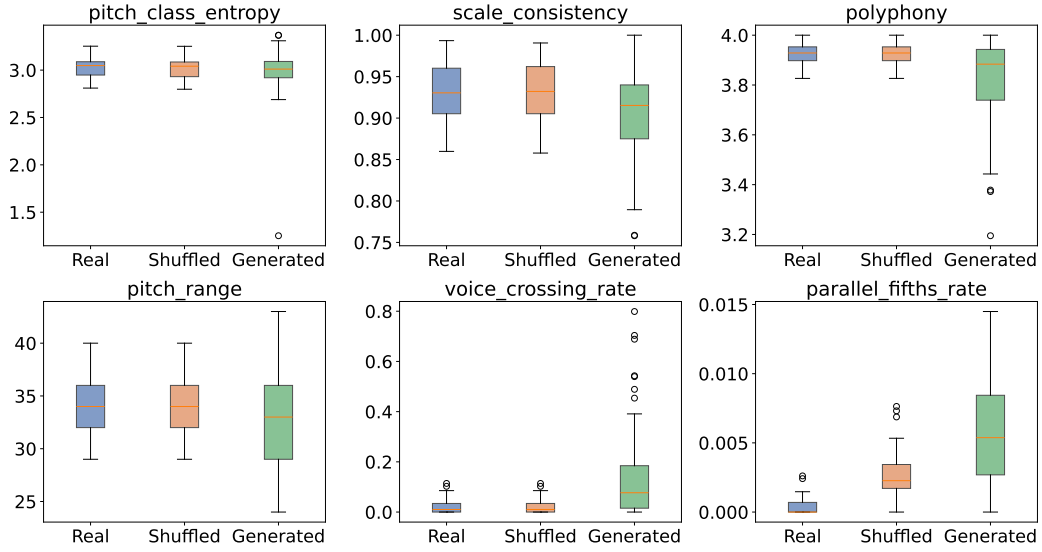


Figure 5: Box plots of the distributional metrics and the rule-based voice crossing and parallel fifths/octaves metrics, applied on real Bach, shuffled Bach and generated chorales. The generated chorales set produces a notably wider range of results than the real Bach and shuffled sets. The set of shuffled chorales is only visibly different for the parallel fifths/octaves metric.

Table 1: Evaluation results of all tested metrics on the real Bach, shuffled Bach and the generated chorales datasets. For the segment-boundary intervals, per-chorale means are used; two generated chorales were excluded because their length was less than a single bar. Model likelihood is not computed for generated chorales, as the model trivially assigns high likelihood to its own samples.

Metric	Real Bach	Shuffled	Generated
<i>Distributional (MusPy)</i>			
Pitch class entropy	3.026	3.021	2.985
Scale consistency	0.930	0.930	0.903
Pitch range (semitones)	34.351	34.351	32.610
Polyphony	3.923	3.923	3.817
<i>Rule-based</i>			
Voice crossing rate	0.021	0.021	0.142
Parallel fifths/octaves rate	0.0004	0.0026	0.0059
Proper cadence	75/77 (97.4%)	62/77 (80.5%)	17/77 (22.1%)
<i>Segment-boundary intervals</i>			
Within-segment	0.342	0.342	0.457
At boundary	1.829	3.595	2.139
Boundary/within ratio	5.35	10.52	4.68
<i>Model likelihood</i>			
NLL	0.558	0.617	–
PPL	1.748	1.853	–

Table 2: Statistical significance and effect sizes of the differences between the real Bach chorales and the shuffled and generated sets. Mann–Whitney U tests are applied for continuous metrics, and Fisher’s exact test for the binary cadence metric. Effect sizes are reported as the absolute rank-biserial correlation $|r_{rb}|$. Unadjusted p -values are shown; results significant after Holm–Bonferroni correction ($p_{HB} < 0.05$) are in bold. Only the within-segment interval (real vs. generated) loses significance after correction.

Metric	Real vs. Shuffled		Real vs. Generated	
	p	$ r_{rb} $	p	$ r_{rb} $
<i>Mann–Whitney U</i>				
Pitch class entropy	0.817	0.022	0.249	0.108
Scale consistency	0.868	0.016	0.002	0.290
Polyphony	1.000	0.000	< 0.001	0.395
Pitch range	1.000	0.000	0.004	0.269
Voice crossing rate	1.000	0.000	< 0.001	0.492
Parallel fifths rate	< 0.001	0.846	< 0.001	0.848
Boundary interval	< 0.001	0.840	0.355	0.087
Within-segment interval	1.000	0.000	0.025	0.210
<i>Fisher’s exact</i>				
Cadence quality	0.001	–	< 0.001	–

5.2 Results

The generated chorales were notably shorter than the real Bach chorales, as illustrated in Figure 4. This may be attributed to over-representation of the EOS token in the training data, as a result of EOS padding that was added to keep all chunks the same length. One of the 77 generated chorales did not produce an EOS token within the maximum generation length. The results of applying all metrics to the three test sets can be viewed in Table 1. The statistical evaluation of the results with p -values and effect sizes can be found in Table 2.

All metrics except pitch class entropy and the segment-boundary interval analysis showed a statistically significant difference between the real Bach and generated sets. The distributional metrics show more overlap between results than rule-based metrics voice crossing and parallel fifths, as visualized in Figure 5, as well as illustrated by the differences in effect size shown in Table 2.

Among the distributional metrics, pitch range and polyphony are identical ($p = 1.000$) for the real Bach and shuffled sets, and nearly identical for pitch class entropy and scale consistency. For the latter, a small effect is expected because shuffling segments only changes local note content when a note held across a segment boundary is split, and none of the distributional metrics measure continuity.

Voice crossing and parallel fifths show the strongest effect between real and generated sets ($r = 0.49$ and 0.85 , respectively), while boundary interval and parallel fifths show the strongest effect between real and shuffled sets ($r = 0.84$ and 0.85 , respectively). While no rank-biserial correlation can be calculated for the cadence quality, there is a notable jump from 97.4% to 22.1% between the real and generated sets. Note that the cadence metric for the shuffled set is attenuated because any remainder segments are kept at the end during shuffling, preserving the original cadence in some chorales. Despite this, the metric still detects a significant difference.

In the segment-boundary analysis, the shuffled set scores substantially higher on boundary intervals than both other sets while within-segment intervals remain identical to real Bach. This confirms that shuffling the sets only disrupts continuity at segment boundaries. By contrast, the generated set remains statistically indistinguishable from real Bach at boundaries.

5.3 Interpretation and positioning

None of the distributional metrics can differentiate between real Bach and shuffled Bach, indicating that these metrics cannot detect the sequential structure of music. Scale consistency, polyphony and pitch range did exhibit statistically significant results when distinguishing real Bach from the generated chorales set. However, each of these results had rank-biserial correlation below 0.4, which indicates that at least 30% of the time, a generated chorale would score equal or higher than a real one, as visualized in Figure 5. So while these metrics are effective at detecting distributional drift, they cannot capture individual quality of pieces. A metric that measures musical quality should arguably be capable of both.

Out of the rule-based metrics, the parallel fifths rate was particularly effective, as it was able to distinguish both the shuffled and generated chorales from real Bach with the strongest effect size of all metrics. At $r = 0.85$, in a randomly drawn real-generated pair, the generated chorale exhibits a higher parallel fifths rate roughly 93% of the time. Because this metric is dependent on temporal behavior, it can capture both local and structural incoherence. By comparison, none of the standard MusPy metrics take into account behavior over sequential timesteps. The high discriminative power of this metric suggests that parallel fifths arise by chance without a deliberate compositional effort to suppress them. This does limit the metric’s applicability to traditions in which parallel fifths are actively avoided, limiting its generalizability. From this it could be argued that the best metrics are based on aspects highly specific to the compositional style that is being assessed.

Similarly, the other metrics display even more distinct blind spots. The voice crossing metric can distinguish between real Bach and the generated set with a modest discriminative power ($r = 0.49$, ~75% identification), but is identical for real and shuffled Bach. Because shuffling preserves the SATB pitch ordering within each segment, this metric is blind to structural disruption. Conversely, the segment-boundary interval analysis detects the shuffled set ($r = 0.84$), but not the generated set, suggesting that the generated chorales have a comparable melodic continuity at segment boundaries to real Bach chorales. Notably, the generated boundary ratio (4.68) is below real Bach’s (5.35),

which would naively rank the flawed model above Bach if smaller jumps are regarded as more coherent. The cadence quality metric is effective in distinguishing real Bach from generated, but it is limited in applicability in that it can only be tested on the end of music pieces.

The increase in NLL between the real Bach and shuffled sets is modest, but consistent with the expectation that shuffling disrupts sequential structure. Because NLL is an aggregate value, no significance test is performed on it, but the difference suggests that the model learned sequential structure. NLL has its own limitation distinct from the other metrics: it is unable to evaluate the model’s own output.

These results demonstrate that the discriminative power of each metric is confined only to a limited domain. If a set of metrics is unable to reliably assess musical quality, published results using them can overstate the actual capabilities of models. This introduces integrity and transparency issues, as this can be misleading to anyone relying on these claims, such as other researchers and artists that may be using or competing with these models. A model evaluated with inadequate metrics could be published as “comparable to Bach” on the basis of scores that do not support that claim. Given the limitations of each metric, researchers should give due consideration to their usage and always be transparent about their limitations.

These findings provide a mechanism for the discrepancies between objective metrics and human evaluation documented in the literature. In the MusicGen paper, Copet et al. [2023] noted that their smallest model scored best when evaluated with the FAD metric, but worst with human evaluators. FAD is a distributional metric, as it measures summary statistics (the mean and covariance of embeddings) [Kilgour et al., 2019]. While MusicGen is a waveform-based model, this discrepancy is consistent with the results of distributional metrics presented here, which suggest that aggregate statistical comparisons can miss structural quality. Likewise, in the NotaGen paper, Wang et al. [2025] found that PPL worsened after optimization while scoring higher on human evaluation. The NLL results presented here only increase modestly for shuffled Bach, despite obvious structural disruption, which implies a discrepancy similar in nature to what Wang et al. described.

6 Conclusions and Discussion

6.1 Findings

Distributional metrics cannot distinguish real Bach from the shuffled set, and have a relatively small effect size when comparing real Bach to the generated set. They are therefore only suitable for evaluating sets as a whole, and only on non-sequential qualities. Rule-based metrics, by contrast, show strong discriminative power, but are genre-dependent. In addition, as each metric is blind to failures outside its own domain, single-metric evaluation is insufficient for assessing musical quality.

6.2 Broader context

In music generation literature, evaluation remains a limiting factor. An advantage of utilizing the symbolic representation of audio data is that it makes rule-based analysis possible: counterpoint conventions can be directly verified on symbolic tokens. This is an option that waveform-based models lack, giving symbolic representations a distinct advantage for evaluation transparency.

Although what constitutes high quality music is inherently subjective, Huron [2001] proposes that there are distinct factors that contribute to whether voice-leading (the practice of connecting chords through individual melodic lines) may be experienced as pleasing. One of these factors is perceptual clarity, of which avoiding parallel fifths is one of many components. As such, while rule-based metrics may measure real indicators of musical quality, each only captures one narrow dimension of quality.

Generative symbolic music models may benefit from being trained on rule-based metrics corresponding to a particular style of music. For example, by minimizing the parallel fifths metric it may learn to approximate Bach’s compositional style more closely. However, metrics used for optimization can no longer be used as independent evaluation measures. This has long been established as a known pitfall in various fields involving statistics, perhaps most famously under Goodhart’s Law [Goodhart, 1975], which states that “*any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes.*” This has been noted in sociology as Campbell’s

Law, which observes that the more any social indicator is used for decision-making, the more apt it will be to distort the very processes it is intended to monitor [Campbell, 1979], as well as in AI alignment literature, as overoptimization can satisfy a metric without achieving the underlying goal it was meant to indicate [Manheim and Garrabrant, 2019]. Because generative music models generally share the same training paradigm as large language models, the metric blindness documented here may extend beyond music. In any domain that evaluates autoregressive output, single-metric evaluation carries the same risks.

An ethical concern with using rule-based metrics is that, when more generally applied, this would reinforce the common practices of one particular tradition over others e.g. scale consistency assumes Western tonal music. This may marginalize cultural frameworks that do not follow those rules. This highlights the subjective nature of music, and the importance of evaluating different compositional styles with their own customized rules.

6.3 Limitations and future work

A limitation of this study is that only one single-seed training run was performed. Multiple training runs may have given more insight into PPL variability. Furthermore, the dataset size of 77 samples per set is relatively small, limiting the statistical power. While using a weak model to generate the chorales accentuated the difference between metrics, this contrasts with popular symbolic music generation models, which are trained on large datasets that span many genres. Results obtained with a weaker model may not necessarily confirm whether the same blind spots persist with stronger models. Moreover, with stronger models there may have been even more overlap between results from the distributional metrics. Moreover, no human evaluation was performed. The results would benefit from a follow-up study that correlates these metric families with the judgments of human listeners.

To investigate the evaluation of musical quality further, a stronger, more generalized model could be trained on a wide range of compositional styles of music across cultures. Through the analysis of such a model, e.g. through t-Distributed Stochastic Neighbor Embedding (t-SNE), cross-cultural markers of musical quality could be searched for. For instance, through t-SNE analysis on NotaGen, Wang et al. [2025] demonstrated distinct label groupings on composers, time periods and instrumentation. Additionally, generated music from other models could be fed into this model to analyze them through model likelihood, and compared to the results from real music.

Alternatively, a discriminator model could be trained specifically on scoring music based on authenticity by feeding it on both real and generated music from various models, with a focus on penalizing generated music. This would result in a learned metric that may circumvent the blindness of individual metrics through taking into account a more complex range of variables, though such a learned metric would itself be subject to the same overoptimization pressure if adopted as a training target.

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